Lecture 17: Econometrics in the Wild I

Jeremy Majerovitz (University of Notre Dame)

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Econometrics in the Wild

We have now learned many of the key tools used in empirical research in economics

You are now ready to start looking at recent research

Today we will go over a few recent papers

- Focus on examples of instrumental variables

Recap: What are we looking for in IV?

Key assumptions for consistency and asymptotic normality of instrumental variables:

- IID Sampling
- Relevance: $\text{Cov}(X_i, Z_i) \neq 0$
- Orthogonality: $\text{Cov}(\varepsilon_i, Z_i) = 0$
  - As-good-as-random Assignment
  - Exclusion Restriction

For LATE interpretation, we also need monotonicity

As we read these papers, we want to think about:

- Are the necessary assumptions satisfied?
- How should we interpret the results?
  - E.g. Who are the compliers? Can we rule out small/large effects?

The Long-Term Effects of Universal Preschool in Boston (Gray-Lobe, Pathak, and Walters, 2023)

Many economists have hypothesized that there are high returns to educational investment in early childhood

- Randomized evaluations show small-scale, high-quality preschool programs (e.g. Perry Preschool and Carolina Abecedarian) have large benefits later in life

Policymakers investing significant resources to expand availability of preschool

- This involves large-scale programs, like public preschool or federal Head Start
- Evidence for these large-scale programs is more mixed, but mostly not randomized
- This is a very active debate in state politics across the country!

Key Question: What is the long-term effect of large-scale public preschool?

Pre-School Attendance is Endogenous

We are interested in estimating a regression like the following:

$$Y_i = \alpha + \beta \cdot D_i + \varepsilon_i$$

where Y_i is some long-term outcome (e.g. college enrollment) and D_i is an indicator for attending public pre-school

For OLS to be consistent, we would need $\text{Cov}(D_i, \varepsilon_i) = 0$.

Yet in general, we think public pre-school attendance is endogenous.

- Why might $\text{Cov}(D_i, \varepsilon_i) > 0$? How would that bias $\hat{\beta}^{OLS}$?
- Why might $\text{Cov}(D_i, \varepsilon_i) < 0$? How would that bias $\hat{\beta}^{OLS}$?

Gray-Lobe et al. solve endogeneity problem by exploiting Boston's preschool lottery.

Background: Boston Public Preschool Program

Boston Public Schools has operated public preschool programs for decades

Gray-Lobe et al. focus on the city's program for four-year-olds

- Universal Eligibility: All four-year old Boston residents eligible to apply
- Limited Capacity: There were not enough seats for all applicants
- Large-Scale Public Program: Programs are operated in public schools and early learning centers
- Qualified Teachers: Teachers subject to same requirements as BPS public school teachers in higher grades. All had bachelor's and 56% had master's degrees. In comparison, one-third of Head Start teachers have bachelor's degrees.

Analysis focuses on 1997-2003 cohorts: Can see if they attended and graduated college

The Boston Preschool Lottery

Preschool slots are allocated by a centralized school-assignment system

- Families submit an application ranking their preferred preschools
- Applicants get special priority based on geography (schools within walking distance) and sibling status (sibling already goes to that school)
- Algorithm breaks ties with a random number generator

Assignment thus has a random component but also a non-random component

- E.g. students who live close to a preschool are more likely to get in

Empirical Strategy

The authors have the full set of applications and know the assignment algorithm

Let Z_i be an indicator for receiving a public preschool offer

Simulate assignment and compute $P_i = P\left(Z_i = 1 \mid \text{preferences and priorities}\right)$

Run regressions on the sample "randomized applicants:" those for whom $0 < P_i < 1$

Main regression equation:

$$Y_i = \beta \cdot D_i + \sum_p \alpha_p 1\left(P_i = p\right) + \text{Controls} + \varepsilon_i$$

where $1\left(P_i = p\right)$ are dummy variables for each possible value of P_i.

Why do we need to control for these dummy variables?

Review Questions

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Instrument: Z_i =Indicator equal to 1 if i receives an offer

- What is the first-stage equation?

Review Questions

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- What is the reduced-form equation?

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- What is the first-stage equation?
- What is the reduced-form equation?
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- Suppose the mayor of Boston is considering expanding the number of preschool slots. Is the LATE from this instrument useful?

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- What is the first-stage equation?
- What is the reduced-form equation?
- Who are the compliers? What LATE does the IV estimate?
- Suppose the mayor of Boston is considering expanding the number of preschool slots. Is the LATE from this instrument useful?
- Suppose the mayor of Boston is considering making preschool mandatory. Is the LATE from this instrument useful?

Results

For the headline result, Y_i is an indicator for on-time college enrollment

- First stage: $\hat{\pi} = 0.645, \text{SE} = 0.015$
- Reduced Form: $\hat{\rho} = 0.054, \text{SE} = 0.019$
- IV Estimate: $\hat{\beta} = 0.083, \text{SE} = 0.030$
- Non-offered Mean: $\mathbb{E}[Y_i \mid Z_i = 0] = 0.459$

How do we test for a weak instrument? Is the instrument strong?

Is the IV estimate statistically significant at the 5% level?

What does a β of 0.083 mean? Is that large or small?

Results

TABLE III

EFFECTS OF PRESCHOOL ATTENDANCE ON POSTSECONDARY OUTCOMES

	Enrollment on time				Enrollment at any time			
	Nonoffered mean (1)	First stage (2)	Reduced form (3)	2SLS estimate (4)	Nonoffered mean (5)	First stage (6)	Reduced form (7)	2SLS estimate (8)
Any college	0.459	0.645***	0.054***	0.083***	0.650	0.645***	0.035*	0.054*
		(0.015)	(0.019)	(0.030)		(0.015)	(0.019)	(0.029)
	2,669		4,175		2,669		4,175	
Two-year college	0.096		−0.002	− 0.003	0.291		0.019	0.030
			(0.011)	(0.017)			(0.018)	(0.028)
	2,669		4,175		2,669		4,175	
Four-year college	0.367		0.056***	0.086***	0.506		0.038*	0.059*
			(0.019)	(0.030)			(0.020)	(0.030)
	2,669		4,175		2,669		4,175	
Massachusetts college	0.328		0.056***	0.087***	0.504		0.045**	0.071**
			(0.019)	(0.029)			(0.020)	(0.030)
	2,669		4,175		2,669		4,175	
Public college	0.257		0.017	0.027	0.474		0.033*	0.051*
			(0.018)	(0.027)			(0.020)	(0.031)
	2,669		4,175		2,669		4,175	

Additional Results

Also boosts high school graduation and reduces disciplinary issues

No significant effect on college graduation (but can't rule out modest effects)

No significant effect on state test scores (test scores start in grade 3)

Consistent with a common finding elsewhere in the literature:

- Early childhood interventions often do not have durable effects on test scores
- But persistent effects on disciplinary outcomes
- Education effects "reappear" later (e.g. college attendance)
- This sort of pattern is sometimes attributed to "non-cognitive skills"

Heterogeneity analysis: Effects seem to be larger for boys than girls

Eviction and Poverty in American Cities (Collinson et al., 2024)

Housing instability is a potentially important driver of poverty

- Losing your home is very disruptive: hard to keep a job and stay safe/healthy

Yet poverty is a major driver of housing instability

- Hard to disentangle causality from reverse causality!

Key Question: What is the causal effect of eviction on subsequent economic outcomes?

Collinson et al. (2024) study this in Cook County (includes Chicago) and New York City

- Exploit random assignment of housing court judges

If you are interested in these topics, Prof. Collinson teaches a course on the "Economics of Housing"

Eviction is Endogenous

Suppose we are interested in the causal effect of eviction

$$Y_i = \alpha + \beta \cdot E_i + \varepsilon_i$$

where E_i is an indicator for eviction, and Y_i is an economic outcome (e.g. earnings)

In general, we think eviction is endogenous.

- Why might $\text{Cov}(E_i, \varepsilon_i) \neq 0$? What direction do you expect the bias to go?

Collinson et al. solve endogeneity problem by exploiting random assignment of judges to eviction cases.

Background: Eviction Courts

How does the eviction process work?

- Begins when landlord files eviction case against tenant (usually because of unpaid rent)
- Case assigned to a housing court judge, who considers the case
- Judge makes a decision: could be eviction order, no eviction, or some agreement between tenant and landlord
- Note that eviction order does not necessarily actually lead to removal
  - Landlord and tenant may reach new agreement and/or landlord may decide not to enforce it

Judges differ in stringency: some judges are more likely to evict than others

Judge Assignment Process

Which judge you get is partly random.

I will explain the process in Cook County (it is similar in NYC):

- Cook County is divided into six districts, each with own courthouse
- Location of property determines which courthouse the case is assigned to
- Within courthouse, case is randomly assigned to a courtroom
- Judges are assigned to courtrooms on a pre-determined schedule

$\Longrightarrow$ Judge is determined by district + time of filing + random assignment

Judge Stringency Instrument

Let $j(i)$ denote the judge assigned to tenant i

The instrument $Z_{j(i)}$ is the leave-one-out eviction rate of the assigned judge in that year

- For that judge and year, compute the average eviction rate, excluding tenant i
- Measures how stringent the judge is for *other* tenants
- Leave-one-out is common practice in this type of empirical strategy to avoid mechanical correlation

This is called an "examiner design," "judge design," or "leniency design"

- Very popular technique to exploit random component of assignment

Regression Specification

Main equation and first stage:

$$Y_i = \beta \cdot E_i + \text{Court-by-Time Dummies} + \text{Controls} + \varepsilon_i$$
$$E_i = \pi \cdot Z_{j(i)} + \text{Court-by-Time Dummies} + \text{Controls} + u_i$$

Court-by-time dummies are dummy variables for each courthouse in each quarter

Need to control for these because judge assignment is only random conditional on the courthouse your case goes to and the time you file

First Stage

TABLE II
FIRST STAGE

	Cook County		New York	
	(1)	(2)	(3)	(4)
Judge stringency	0.741***	0.740***	0.831***	0.825***
	(0.025)	(0.024)	(0.057)	(0.057)
Controls	No	Yes	No	Yes
Observations	268,000	268,000	150,662	150,662

Question: How can I compute the first-stage F-statistic? Are the instruments weak/strong?

Results

The authors report average effects for Cook County and NYC:

Outcome	IV estimate	Standard error	Nonevicted mean
Emergency shelter use, first year	+0.034	(0.017)	0.009
Average quarterly earnings, first year	−$323	($175)	$4,300
Average quarterly earnings, second year	−$613	($248)	$4,254
Credit score, second year	−16.53	(6.67)	551.84

Eviction raises risk of homelessness, lowers earnings, and worsens credit score

Question: Which of these results is not statistically significant at the 5% level?

Question: Who are the compliers? How might they be different from others?

Instrument Validity and Interpretation

Let's go through our checklist of IV assumptions.

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  - Judges make other decisions besides eviction: Size of monetary penalty, or time until eviction enforced
  - Authors find these are only weakly correlated with stringency
- Monotonicity: What does monotonicity require?

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- Monotonicity: What does monotonicity require?
  - Partial test: Across different groups, first stage is always positive

Do Political Protests Matter? Evidence from the Tea Party Movement (Madestam, Shoag, Veuger, and Yanagizawa-Drott (2013)

Political protests often occur around the time of major political developments

But, not obvious whether the protests cause political change, or just reflect changes that are already happening

Key Question: What effect do political protests have on electoral outcomes?

Madestam et al. will study this question in the context of the Tea Party movement and the 2010 midterm elections

Background: The Tea Party movement and 2010 midterms

The Tea Party movement was a conservative political movement that emerged in early 2009

- Driven by opposition to taxes, government spending, and government intervention/bailouts following the financial crisis
- Named after the Boston Tea Party, when American revolutionaries dumped tea into Boston harbor to protest British taxes on tea

Organized hundreds of Tea Party rallies on April 15, 2009 (Tax Day)

- Estimated that 400,000 to 800,000 protesters attended these rallies
- Tea Party rallies continued afterwards, but these were among the first

In November 2010, Republicans gained 63 House seats and 6 Senate seats, retaking the House

- Swing in seats similar in magnitude to 2026 blue wave

Instrument

Going to a protest is no fun if it rains

Authors use rainfall on April 15, 2009 as an instrument for protest size

$$Z_i = \begin{cases} 1 & \text{if rainfall was at least 0.1 inches} \\ 0 & \text{otherwise} \end{cases}$$

where i indexes counties

Rain is partly random, but rain is much more likely in South Bend, IN than in Phoenix, AZ

- Regressions control for historical rainfall probability in April

This is a *natural experiment*: not a true RCT, but a situation where the world provides something approximating an experiment

Regression Specification

$$Y_i = \alpha + \beta \cdot \text{Protesters}_i + \gamma \cdot \text{Rainfall Probability}_i + \text{Controls} + \varepsilon_i$$

- Y_i is a political outcome, e.g. Republican votes as percentage of county population
- Protesters_i is the percentage of the county population that attended the protests
- $\text{Rainfall Probability}_i$ is the historical probability of rain in April

Results

First-stage coefficient: $\hat{\pi} = -0.082$ percentage points, SE of 0.021 p.p.

- Overall mean attendance was 0.161% of county population
- Rain cut protest size in half
- First-stage F-statistic of about 15: moderately strong but not super-strong

Headline IV result:

- Republican party vote as share of population: $\hat{\beta} = 12.59$, SE=4.21
- Interpretation: One additional protester caused 12.6 additional Republican votes

Thinking About the Orthogonality Condition

We can break the orthogonality condition, $\text{Cov}(Z_i, \varepsilon_i) = 0$ into two parts

As-Good-As-Random Assignment

- This seems quite solid: conditional on historical probabilities, rain is random

Exclusion Restriction

- Rainfall on April 15 must affect political outcome only through effect on number of protesters

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- What if the news doesn't cover the protest when it rains, or if people don't see it?
- What if the quality of the protest is lower because people are having a bad time?
- These would violate the exclusion restriction, and make it hard to interpret the IV estimates.
- However, the authors note that their reduced form estimates would still be valid, and test the hypothesis that the protests mattered for political outcomes.